Agentic AI-Driven Persona-Based Software Removal and Governance System

Abstract

This invention describes an agentic AI-driven system for autonomous software removal and governance that dynamically adapts to user personas, risk contexts, and behavioural patterns.

The system addresses limitations in conventional software management tools that rely on static policies, manual configurations, and limited personalization.

At its core, the invention features a software management agent augmented with an Agentic AI engine capable of real-time risk prediction, context-aware hypothesis testing, and adaptive task scheduling. It incorporates a persona-based action layer that tailors' software removal and application offloading strategies based on user roles, behaviour analytics, and clickstream data.

The system continuously aggregates data from endpoint inventories, browser extensions, executables, scheduled tasks, and service states, feeding into a federated learning framework for decentralized threat intelligence sharing. Unauthorized or underutilized applications are identified, scheduled for removal or offloading, and eliminated using AI-powered residual cleanup mechanisms including registry and file system scans.

Further, the system integrates explainable AI modules for generating persona-specific justifications, approval workflows, and compliance reports, ensuring transparency and auditability. By combining agentic autonomy, personalization, and federated intelligence, this invention delivers a novel approach to software governance that is both proactive and adaptive.

Technical Field

The present invention relates to the field of endpoint software management, specifically to automated systems for software identification, removal, and governance across enterprise environments. More particularly, it pertains to agent-based software removal systems enhanced with artificial intelligence (AI), user persona profiling, adaptive scheduling, and federated threat intelligence mechanisms.

Problem Statement / Background

Traditional software management tools are largely rule-based and operate on static policies that lack personalization and contextual awareness. Existing systems often rely on scheduled scans and manual intervention to identify and remove unauthorized or non-compliant software from user endpoints. These approaches are insufficient in modern enterprise environments where software usage patterns are diverse, dynamic, and role specific.

Conventional removal agents are limited in several ways:

- They do not account for user personas or behavioural data when making removal decisions.
- Software identification is typically based on pre-defined lists, which quickly become outdated.
- Removal actions are performed using fixed schedules, with no regard to user activity, system load, or operational risk.
- Residual files and registry entries are often left behind due to shallow cleanup routines.
- There is minimal support for audit trails, real-time compliance tracking, or transparent justification of automated actions.

Additionally, current tools lack integration with federated learning models, external threat intelligence platforms, and explainable AI systems. As a result, organizations struggle to achieve continuous compliance, optimal resource utilization, and proactive software governance.

The present invention addresses these limitations by introducing an agentic AI-based software removal and governance system that dynamically adapts to each user's environment, behaviour, and risk profile—enabling intelligent, real-time, and auditable software lifecycle management.

Summary of the Solution / Invention

The present invention provides a comprehensive, agent-based software governance system that leverages Agentic Artificial Intelligence (AI), user persona profiling, adaptive scheduling, and federated learning to automate and optimize the software removal process across endpoint devices.

At its core, the system includes a software management agent that continuously collects inventory data—including installed software, browser extensions, executables, scheduled tasks, and system services. This data is aggregated into a central repository and analyzed by an Agentic AI engine comprising several modules: a risk prediction model, an adaptive scheduler, and a hypothesis-generation engine for detecting unauthorized or non-compliant applications.

The system introduces a persona-based action layer, wherein each user endpoint is profiled based on role, behavior, and usage patterns. Removal policies and software offloading decisions are dynamically tailored to each persona, enabling hyper-personalized and minimally disruptive governance. For example, rarely used applications may be offloaded to the cloud or removed entirely during idle periods, depending on the user's work style and risk tolerance.

An execution layer manages task scheduling, software uninstallation, process termination, and advanced residual cleanup—utilizing PowerShell commands and AI-trained models for deep system hygiene. Additionally, the system supports federated learning, enabling anonymized sharing of threat insights across organizations and continuously improving global removal intelligence.

The invention further integrates compliance auditing, explainable AI reports, and real-time dashboards for full transparency, traceability, and control. This novel approach delivers proactive, adaptive, and intelligent software management that is aligned with modern enterprise requirements.

Detailed Description of the Solution Strategy

Overview

The invention presents an autonomous, agentic software governance system designed to intelligently remove or offload unauthorized or underutilized software from endpoint devices. The system is composed of interconnected modules leveraging advanced AI/ML techniques, persona-based intelligence, and federated learning for dynamic and context-aware decision-making.

The architecture comprises the following key components:

1. **Data Collection & Aggregation Layer**
2. **Agentic AI Decision Engine**
3. **Persona-Based Action Layer**
4. **Execution & Enforcement Layer**
5. **Logging, Reporting & Compliance Layer**
6. **Federated Learning & Threat Intelligence Sharing Layer**

Each module interacts with others to form a real-time, adaptive, and explainable software removal framework.

1. Data Collection & Aggregation Layer

This layer performs continuous, structured, and timestamped data collection from user devices. It collects:

- **Software Inventory:** Installed apps, MSI packages, UWP/Store/PWA apps.
- **Executable Inventory:** File paths, signatures, and metadata.
- **Browser Extensions:** Extension IDs, versions, paths, and sources.
- **Scheduled Tasks & Services:** Task scheduler entries and running service states.

- **Hardware Metadata:** CPU, RAM, GPU, BIOS, disk usage, OS version.
- **Behavioral Signals:** Clickstream data, idle time, app usage duration.
- **Threat Feeds:** External threat intelligence from EDR/XDR and SIEM platforms.

The collected data is sent to a central repository through secure API calls and message queues, ensuring synchronization and audit-ready recordkeeping.

2. Agentic AI Decision Engine with Narrow AI

~~This engine is the intelligence core of the system, designed to operate with minimal human intervention by mimicking human-like judgment using AI. It includes:~~

The Agentic AI Engine consists of multiple specialized Narrow AI modules, each trained or programmed for a specific bounded task within the endpoint governance system.
These modules are coordinated by a central AI Decision Controller, which synthesizes module outputs into unified, autonomous, and explainable system actions.

Each module:

- Works independently within its narrow scope,
- Uses specialized models (supervised, unsupervised, or heuristic),
- Is lightweight and interpretable for compliance and auditing.

Key Components:

- ~~**AI Decision Engine:** Orchestrates decision-making based on integrated models and feedback loops.~~
- **Risk Prediction Module:** Assigns a risk score to each software instance based on historical threat intel, behavioral anomalies, and persona-policy deviations.
- **Adaptive Scheduler:** Dynamically adjusts scan intervals and removal windows based on system load, user activity, and threat context.
- **Hypothesis Generation and Testing System (HGTS):** Forms AI-driven assumptions about software legitimacy and tests them against contextual baselines.
- **Context-Aware AI Database (CAAD):** Stores software baselines, removal rules, and adaptive thresholds per persona and organization.

- **Persona Profiler AI**
  Builds and updates dynamic user personas

- **Offloading Decision AI**
  Recommends cloud/offloading actions for low-usage apps

- **Residual Artifact Detector AI**
  Identifies leftover artifacts post-removal

  Example:
  A marketing employee's laptop has Installed apps including Zoom, Photoshop, Torrent client, and several unknown programs.

```
class AgenticAIEngine:
    def __init__(self):
        self.risk_predictor = RiskPredictorAI()
        self.scheduler = AdaptiveSchedulerAI()
        self.hypothesis_tester = HypothesisTestingAI()
        self.persona_profiler = PersonaProfilerAI()
        self.offload_decider = OffloadDecisionAI()
        self.artifact_detector = ResidualArtifactDetectorAI()
        self.decision_controller = AIDecisionController()

    def evaluate_endpoint(self, endpoint_data, user_profile):
        risk_scores = self.risk_predictor.predict(endpoint_data['installed_apps'])
        persona_info = self.persona_profiler.profile(user_profile)

        scan_schedule = self.scheduler.adjust(persona_info,
endpoint_data['system_load'])

        anomalies =
self.hypothesis_tester.detect(endpoint_data['behavioural_logs'])

        offload_recommendations =
self.offload_decider.recommend(endpoint_data['usage_stats'])

        actions = self.decision_controller.synthesize(
            risk_scores=risk_scores,
            anomalies=anomalies,
            persona_info=persona_info,
            offload_recommendations=offload_recommendations
        )

        self.execute_actions(actions)

    def execute_actions(self, actions):
        for action in actions:
            if action['type'] == 'remove':
                self.remove_application(action['app'])
```

```
        elif action['type'] == 'offload':
            self.offload_application(action['app'])
        elif action['type'] == 'notify':
            self.send_notification(action['message'])

    def remove_application(self, app):
        print(f"Removing {app}...")

    def offload_application(self, app):
        print(f"Offloading {app} to cloud storage...")

    def send_notification(self, message):
        print(f"Notification: {message}")

# Example of specialized Narrow AI module
class RiskPredictorAI:
    def predict(self, installed_apps):
        # simple mockup: real version would use ML models
        return {app: "High" if "torrent" in app.lower() else "Low" for app in
installed_apps}

# Controller module
class AIDecisionController:
    def synthesize(self, risk_scores, anomalies, persona_info,
offload_recommendations):
        actions = []
        for app, risk in risk_scores.items():
            if risk == "High":
                actions.append({'type': 'remove', 'app': app})

        for app in offload_recommendations:
            actions.append({'type': 'offload', 'app': app})

        if anomalies:
            actions.append({'type': 'notify', 'message': 'Anomalies detected, admin
notified.'})

        return actions
```

Output:

- **Risk Predictor AI** flags the Torrent client as "High Risk."
- **Persona Profiler AI** knows that marketing personas don't usually need Torrent apps.
- **Adaptive Scheduler AI** increases the scan frequency due to detected anomaly.
- **Hypothesis Testing AI** runs anomaly analysis, confirming unusual download patterns.
- **Offloading Decision AI** suggests offloading a rarely-used Photoshop plugin to cloud storage.

- ➢ **AI Decision Controller** synthesizes all outputs **Auto-Remove** Torrent app, **Offload** Photoshop plugin, **Notify** user and Admin.

AI/ML Models Employed:

- **Supervised Learning (e.g., Random Forests, XGBoost):** Used for classification of software as authorized, unauthorized, or redundant based on labeled training data.
- **Unsupervised Learning (e.g., K-Means Clustering, Isolation Forests):** Detects outliers in software usage patterns or system behavior without predefined labels.
- **Anomaly Detection Models (e.g., Autoencoders, One-Class SVM):** Identify residual artifacts, registry anomalies, or behavioral deviations.
- **Reinforcement Learning:** Fine-tunes scheduling logic and policy decisions over time via reward-feedback loops based on user/system outcomes.
- **Federated Model Aggregation (e.g., Federated Averaging):** Combines decentralized model updates without transferring local data.

These models collectively enable the system to reason, learn, and adapt in real-time, reducing reliance on static rules or hardcoded configurations.

3. Persona-Based Action Layer

This layer ensures that software decisions are not applied uniformly but are instead tailored to user context.

- **Persona Profiling Engine:** Maps each user to a persona (e.g., Developer, Executive, Analyst) using their role, behavior, access level, and software usage history.
- **Custom Policy Generator:** Creates personalized software allowlists/blocklists and removal schedules based on persona-specific rules.
- **Hyper-Personalization Module:** Adapts global persona policies to individual usage patterns—allowing user-specific exceptions or adjusted timing.
- **Application Offloading System:** Flags low-usage apps for removal or offloading to cloud/edge environments, factoring in persona-specific risk levels and storage limits.
- **Approval Workflow Engine:** Automates approval chains for sensitive removal decisions, with optional manual override.

4. Execution & Enforcement Layer

This layer operationalizes the system's decisions through direct actions on user endpoints.

- **Software Management Agent:** Interfaces with task scheduler, removal agent, and cloud services.
- **Task Scheduler Integration:** Configures and triggers timed or event-based removal tasks.
- **Removal Agent:**
  - Terminates unauthorized or flagged processes.
  - Executes PowerShell commands to uninstall apps (MSI, EXE, UWP, portable, PWA).
  - Cleans up registry keys (HKEY_LOCAL_MACHINE, HKEY_CURRENT_USER, etc.).
- **Residual Artifact Detection Module:** Uses AI-driven heuristics and anomaly models to detect and eliminate hidden leftovers (e.g., registry paths, temp files).
- **Offload Agent:** Moves identified low-priority applications to secondary storage or a remote location for space optimization.

5. Logging, Reporting & Compliance Layer

Every action, decision, and outcome is tracked and reportable:

- **Status Logging Module:** Logs success/failure of removal tasks, timestamps, and approval status.
- **Explainable AI Report Generator:** Outputs plain-language justifications per persona and per removal action.
- **Compliance & Audit Reports:** Tracks policy adherence, license usage, and organizational risk exposure.
- **Real-Time Dashboards & Alerts:** Allow administrators and users to monitor system actions and receive relevant notifications.

6. Federated Learning & Threat Intelligence Sharing Layer

To enhance collective intelligence across organizations while preserving data privacy:

- **Federated Learning Coordinator:** Aggregates anonymized local model updates to improve global AI model accuracy.
- **Community Intelligence Sharing:** Shares flagged application fingerprints, suspicious behavior patterns, and threat scores.

- **Retraining Pipeline:** Periodically refreshes the central CAAD and risk models based on feedback from multiple deployments.

Operational Workflow (Summary)

1. **Inventory Collection:** The agent collects detailed endpoint and software data.
2. **AI Processing:** Data is fed into the decision engine, which determines risk, schedule, and persona impact.
3. **Persona Policy Application:** Software is evaluated against persona-aligned rules and workflows.
4. **Task Execution:** Removals, offloads, and cleanups are triggered via the execution layer.
5. **Logging & Feedback:** Actions are logged, reported, and used to refine future decisions via learning loops.

Claims

Claim 1 (Independent Claim – System-Level)

1. A system for autonomous software removal and governance, comprising:
   a. a data collection agent configured to aggregate software inventory, executable metadata, browser extensions, system tasks, and hardware profiles from user endpoints;
   b. an Agentic AI decision engine comprising:
      i. a risk prediction module utilizing supervised and unsupervised machine learning models to classify and score software;
      ii. a hypothesis generation and testing module for detecting unauthorized or redundant applications based on behavioral patterns and historical data;
      iii. an adaptive scheduler that dynamically adjusts scanning and removal frequencies based on risk score, user activity, and system context;
   c. a persona profiling module configured to determine user roles and behavior from application usage data, clickstream analytics, and system interactions;
   d. a software policy engine that generates customized removal or offloading rules based on persona type;
   e. a removal execution engine configured to:
      i. terminate flagged processes;
      ii. uninstall applications via command-line execution; and
      iii. perform registry and file cleanup using AI-based residual detection;
   f. a compliance and logging system configured to record decisions, generate explainable justifications, and create audit reports; and

g. a federated learning engine configured to aggregate anonymized removal insights from distributed instances to update global AI models.

Claim 2 (Dependent – Model Details)

2. The system of claim 1, wherein the risk prediction module employs one or more machine learning models selected from the group consisting of: random forest classifiers, XGBoost models, k-means clustering, isolation forests, and autoencoders.

Claim 3 (Dependent – Persona Logic)

3. The system of claim 1, wherein the persona profiling module assigns users to persona groups based on historical application usage frequency, idle time, interaction patterns, and access control metadata.

Claim 4 (Dependent – Scheduling Intelligence)

4. The system of claim 1, wherein the adaptive scheduler is further configured to modify task execution windows based on CPU load, user availability, and historical task success rates.

Claim 5 (Dependent – Removal Agent Capabilities)

5. The system of claim 1, wherein the removal execution engine supports uninstallation of applications including MSI packages, non-MSI executables, portable apps, UWP apps, and progressive web applications.

Claim 6 (Dependent – Artifact Detection)

6. The system of claim 1, wherein the registry and file cleanup utilizes anomaly detection models trained to identify unauthorized residual software artifacts.

Claim 7 (Dependent – Approval Workflows)

7. The system of claim 1, further comprising an approval workflow module that automates user or admin confirmations prior to executing software removal actions.

Claim 8 (Dependent – Offloading Mechanism)

8. The system of claim 1, wherein the software policy engine is further configured to offload low-usage applications to cloud or secondary storage based on persona-specific thresholds.

Claim 9 (Dependent – Explainability Layer)

9. The system of claim 1, wherein the compliance and logging system includes an explainable AI module that generates human-readable reasons for each removal decision, specific to the affected persona.

Claim 10 (Dependent – Federated Intelligence)

10. The system of claim 1, wherein the federated learning engine utilizes federated averaging to incorporate distributed model updates from multiple endpoints without transferring raw user data.

Patent Drawings & Descriptions

FIG. 1 – Data Collection & Aggregation Layer

Description:
FIG. 1 illustrates the data collection and aggregation layer of the system. This layer comprises multiple subcomponents responsible for gathering endpoint and contextual information, including:

- 102: Software Inventory Collector – collects details of installed software and applications.
- 104: Hardware Inventory Collector – gathers device-specific hardware metrics.
- 106: Behavioural Tracker – monitors clickstream data and software usage behaviour.
- 108: Threat Intelligence Integrator – interfaces with external EDR/XDR or threat feeds.
- 110: Central Repository – stores the collected data for downstream analysis.

Each component forwards data to the central repository for unified analysis and AI processing.

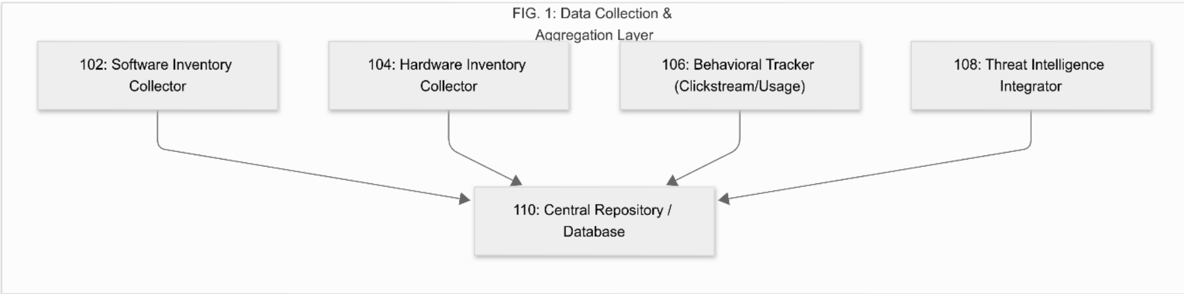


FIG. 2 – Agentic AI Decision Engine

Description:
FIG. 2 shows the Agentic AI Engine that powers autonomous decision-making within the system. It includes:

- 202: Risk Prediction Module – assigns risk scores using supervised and unsupervised ML models.
- 204: Adaptive Scheduler – adjusts scanning/removal intervals based on context and load.
- 206: Hypothesis Generation and Testing System (HGTS) – detects anomalous applications and behaviours.
- 208: Context-Aware AI Database (CAAD) – stores historical patterns, policies, and thresholds.
- 210: AI Decision Controller – coordinates outputs from various AI submodules.

This engine enables intelligent classification, action planning, and adaptive system responses.

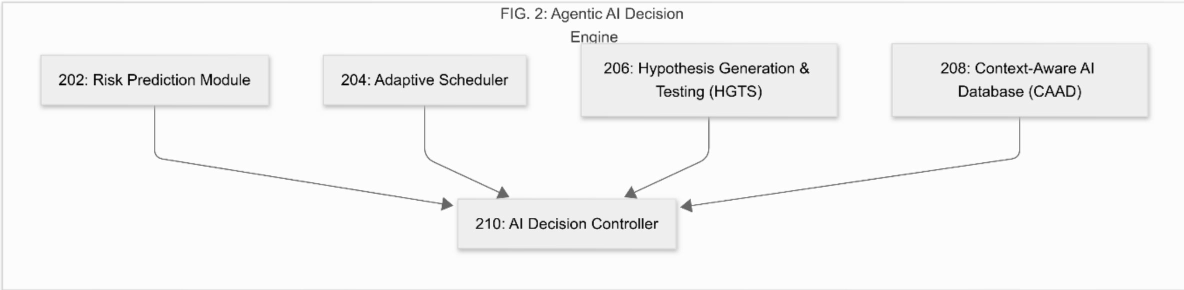


FIG. 3 – Persona-Based Action Layer

Description:
FIG. 3 illustrates the persona-based action logic used to tailor decisions according to user roles and behaviours. It includes:

- 302: Persona Profiler – builds dynamic user profiles based on activity, usage, and access level.
- 304: Policy Generator – generates persona-aligned removal and offloading policies.
- 306: Hyper-Personalization Engine – refines policy behavior at the individual level.
- 308: Approval Workflow Module – integrates human approvals into automation logic.
- 310: Application Offloading Controller – manages the transfer of rarely-used apps to secondary or cloud storage.

These components collectively personalize system behaviour for each user type.

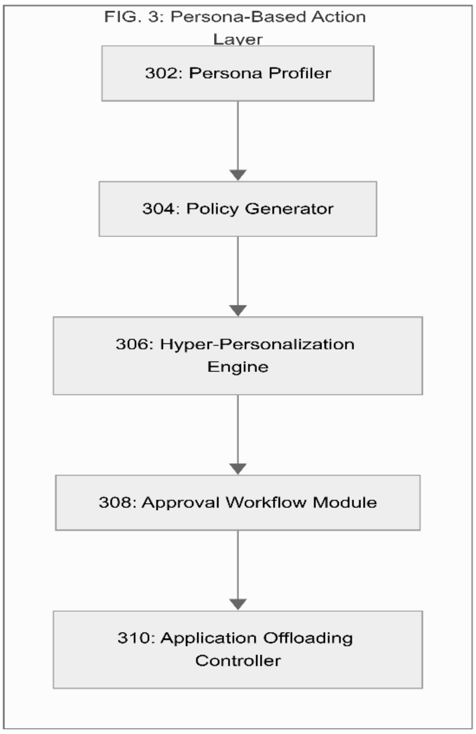


FIG. 4 – Execution & Enforcement Layer

Description:
FIG. 4 depicts the system's operational engine responsible for executing removal and offloading actions. Components include:

- 402: Software Management Agent – orchestrates overall task execution.
- 404: Task Scheduler Interface – interfaces with OS-level scheduling systems.
- 406: Removal Agent – performs uninstallation via PowerShell, command-line, and custom scripts.
- 408: Registry & File Cleaner – cleans up software traces post-removal.
- 410: Offload Agent – manages cloud or network offloading of flagged applications.
- 412: Residual Artifact Detection Module – uses AI to detect hidden or leftover system artifacts.

Together, these elements ensure reliable and complete software remediation.

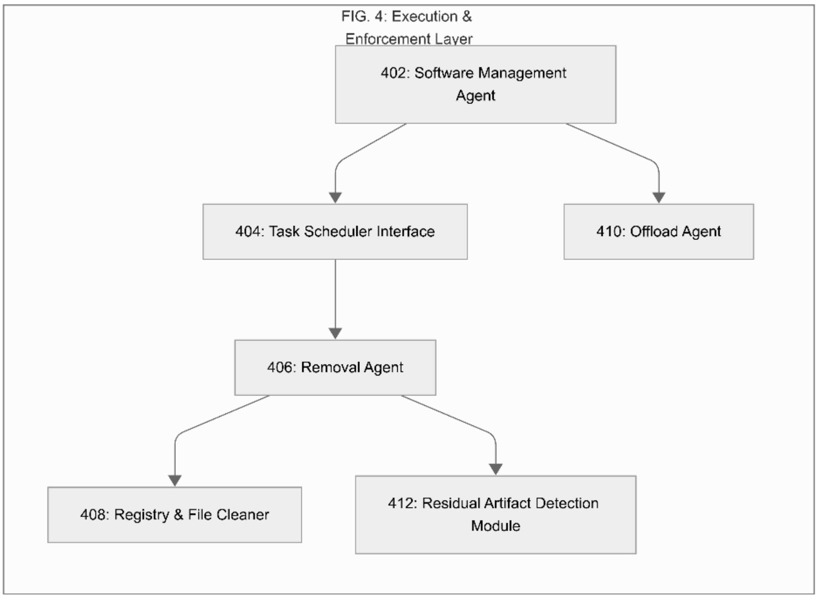


FIG. 5 – Logging, Reporting & Compliance Layer

Description:
FIG. 5 shows the components responsible for transparency, traceability, and compliance. It includes:

- 502: Status Logger – logs all removal, scheduling, and offloading events.
- 504: Explainable AI Report Generator – provides human-readable reasons for AI actions.
- 506: Compliance Report Generator – supports audits and policy alignment.
- 508: Notification & Dashboard Interface – provides real-time feedback to users and administrators.

These components create a robust accountability framework for the software governance system.

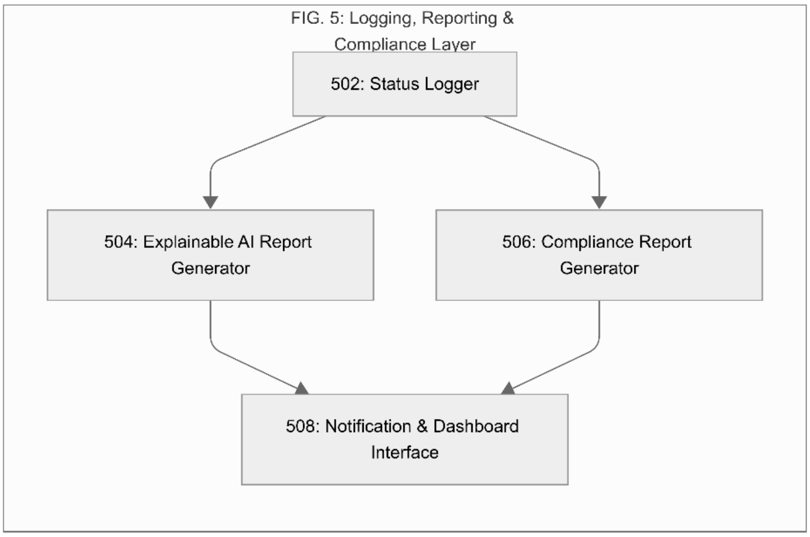


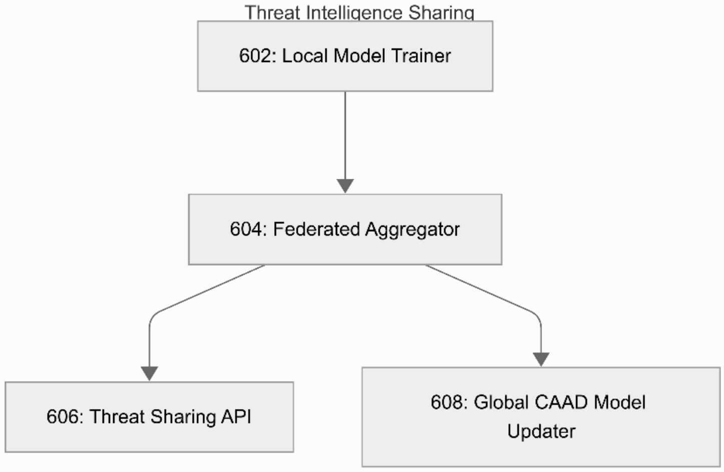
FIG. 6 – Federated Learning & Threat Intelligence Sharing

Description:
FIG. 6 illustrates the federated learning and external intelligence integration components, including:

- 602: Local Model Trainer – trains local models on anonymized endpoint data.
- 604: Federated Aggregator – aggregates model updates across instances.
- 606: Threat Sharing API – enables anonymized sharing of flagged software fingerprints and behaviour.
- 608: Global CAAD Model Updater – updates the central knowledge base with distributed insights.

This subsystem allows the invention to evolve collaboratively across organizations while preserving data privacy.

FIG. 6: Federated Learning & Threat Intelligence Sharing



Strategic Benefits to Dell Technologies

Alignment with Enterprise Strategy and Innovation Roadmap

The proposed invention—*Agentic AI-Driven Persona-Based Software Removal and Governance System*—delivers substantial strategic value to Dell Technologies by addressing emerging enterprise challenges in intelligent endpoint governance, compliance automation, and user-centric device management.

This system exemplifies Dell's commitment to intelligent automation, security-first design, and adaptive technologies. It enhances Dell's software portfolio across commercial device management, client security, and digital workspace optimization by offering:

- Autonomous Device Hygiene:
  The invention supports zero-touch endpoint management by autonomously detecting, removing, or offloading unauthorized or redundant software. This enhances Dell's offerings in scalable device fleet management and

complements solutions like Dell Client Command Suite and Unified Workspace.

- AI-Augmented Endpoint Security:
  By integrating supervised and unsupervised learning models with federated intelligence, the system strengthens Dell's endpoint security posture. It positions Dell as a thought leader in AI-driven threat response, compliance enforcement, and behavioural anomaly detection.
- Persona-Centric User Experience:
  Through hyper-personalization, the system aligns with Dell's focus on enhancing user productivity and satisfaction. It intelligently preserves critical applications while reducing clutter and risk, reinforcing Dell's differentiated approach to tailored enterprise experiences.
- Market Differentiation & IP Strengthening:
  The invention introduces novel mechanisms—including Agentic AI, adaptive scheduling, federated learning, and explainable remediation—which distinguish Dell from traditional software management vendors. Securing IP protection over these technologies further strengthens Dell's defensible innovation portfolio.
- Cross-Solution Synergies:
  This system can integrate seamlessly with existing Dell offerings such as Dell Optimizer, Data Security, Workspace ONE, and telemetry-driven cloud services—expanding use cases across commercial, education, and federal markets.